

1. Evaluate the following integrals by applying the formulas in Lecture 1.5 Example 6.

$$1). \int \cos(4x) dx = \frac{1}{4} \sin(4x) + C$$

$$2). \int e^{-2x} dx = -\frac{1}{2} e^{-2x} + C$$

$$3). \int \sin\left(\frac{x}{2}\right) dx = -2 \cos\left(\frac{x}{2}\right) + C$$

$$4). \int \frac{1}{4+3x} dx = \frac{1}{3} \ln|4+3x| + C$$

2. (1.5) Evaluate the following integrals by substitutions. Make sure to indicate the u -function.

$$1). \int x^2 \sqrt{x^3+1} dx \quad u = x^3+1 \\ du = 3x^2 dx$$

$$= \int \sqrt{u} \cdot \frac{1}{3} du$$

$$= \frac{1}{3} \cdot \frac{2}{5} u^{5/2} + C$$

$$= \frac{2}{9} (x^3+1)^{5/2} + C$$

$$2). \int \frac{\sec^2(1/x)}{x^2} dx \quad u = \frac{1}{x} \Rightarrow du = -\frac{1}{x^2} dx$$

$$= - \int \sec^2 u du$$

$$= -\tan u + C$$

$$= -\tan\left(\frac{1}{x}\right) + C$$

$$3). \int e^x \sin(e^x) dx \quad u = e^x \\ du = e^x dx$$

$$= \int \sin u du$$

$$= -\cos u + C$$

$$= -\cos(e^x) + C$$

$$4). \int \frac{x}{x^2+1} dx \quad u = x^2+1 \\ du = 2x dx$$

$$= \frac{1}{2} \int \frac{1}{u} du$$

$$= \frac{1}{2} \ln|u| + C$$

$$= \frac{1}{2} \ln(x^2+1) + C$$

↓
(we don't need the absolute value sign b/c $x^2+1 > 0$)

$$u = 1 + \tan \theta$$

$$du = \sec^2 \theta d\theta$$

$$5). \int (1 + \tan \theta)^5 \sec^2 \theta d\theta$$

$$= \int u^5 du$$

$$= \frac{1}{6} u^6 + C$$

$$= \frac{1}{6} (1 + \tan \theta)^6 + C$$

$$6). \int \frac{e^x}{e^x + 1} dx$$

$$= \int \frac{1}{u} du$$

$$u = e^x + 1$$

$$du = e^x dx$$

$$= \ln |u| + C$$

$$= \ln |e^x + 1| + C$$

$$7). \int \frac{\sin x}{1 + \cos^2 x} dx$$

$$u = \cos x$$

$$du = -\sin x dx$$

$$= - \int \frac{1}{1 + u^2} du$$

$$= -\tan^{-1} u + C$$

$$= -\tan^{-1} (\cos x) + C$$

$$8). \int x^5 (x^6 + 10)^{10} dx$$

$$u = x^6 + 10$$

$$du = 6x^5 dx \Rightarrow x^5 dx = \frac{1}{6} du$$

$$= \int u^{10} \cdot \frac{1}{6} du$$

$$= \frac{1}{6} \cdot \frac{1}{11} u^{11} + C$$

$$= \frac{1}{66} (x^6 + 10)^{11} + C$$

$$9). \int \frac{\cos \theta}{\sin^3 \theta} d\theta \quad \begin{array}{l} u = \sin \theta \\ du = \cos \theta d\theta \end{array}$$

$$= \int \frac{1}{u^3} du$$

$$= \int u^{-3} du$$

$$= -\frac{1}{2} u^{-2} + C$$

$$= -\frac{1}{2} \cdot (\sin \theta)^{-2} + C$$

$$= -\frac{1}{2} \frac{1}{\sin^2 \theta} + C$$

$$10). \int \frac{1}{x \ln x} dx \quad \begin{array}{l} u = \ln x \\ du = \frac{1}{x} dx \end{array}$$

$$= \int \frac{1}{u} du$$

$$= \ln |u| + C$$

$$= \ln |\ln x| + C$$

$$11). \int \sin(\tan \theta) \sec^2 \theta d\theta \quad \begin{array}{l} u = \tan \theta \\ du = \sec^2 \theta d\theta \end{array}$$

$$= \int \sin u du$$

$$= -\cos u + C$$

$$= -\cos(\tan \theta) + C$$

$$12). \int_0^1 \frac{(\arctan x)^2}{1+x^2} dx$$

$$\begin{array}{l} u = \arctan x \\ du = \frac{1}{1+x^2} dx \end{array}$$

$$\text{when } x=0: \quad u = \arctan 0 = 0$$

$$x=1: \quad u = \arctan 1 = \frac{\pi}{4}$$

$$= \int_0^{\frac{\pi}{4}} u^2 du$$

$$= \frac{1}{3} u^3 \Big|_0^{\frac{\pi}{4}}$$

$$= \frac{1}{3} \frac{\pi^3}{4^3} - 0$$

$$= \frac{\pi^3}{192}$$

$$\begin{aligned}
 13). \int_0^{\pi/3} \sec^3 \theta \tan \theta \, d\theta & \quad \begin{array}{l} u = \sec \theta \\ du = \sec \theta \tan \theta \\ \text{when } \theta = 0: u = 1 \\ \text{when } \theta = \frac{\pi}{3}: u = 2 \end{array} \\
 &= \int_1^2 u^2 \, du \\
 &= \left. \frac{1}{3} u^3 \right|_1^2 \\
 &= \frac{8}{3} - \frac{1}{3} \\
 &= \frac{7}{3}
 \end{aligned}$$

$$\begin{aligned}
 14). \int_1^2 x \sqrt{x-1} \, dx. & \quad \begin{array}{l} u = x-1 \Rightarrow x = u+1 \\ du = dx \\ \text{when } x=1: u=0 \\ \text{when } x=2: u=1 \end{array} \\
 &= \int_0^1 (u+1) \sqrt{u} \, du \\
 &= \int_0^1 (u^{3/2} + u^{1/2}) \, du \\
 &= \left(\frac{2}{5} u^{5/2} + \frac{2}{3} u^{3/2} \right) \Big|_0^1 \\
 &= \frac{2}{5} + \frac{2}{3} \\
 &= \frac{6}{15} + \frac{10}{15} \\
 &= \frac{16}{15}
 \end{aligned}$$

$$\begin{aligned}
 15). \int x^3 (2+x^4)^5 \, dx & \quad \begin{array}{l} u = 2+x^4 \\ du = 4x^3 \, dx \\ (\Rightarrow \frac{1}{4} du = x^3 \, dx) \end{array} \\
 &= \frac{1}{4} \int u^5 \, du \\
 &= \frac{1}{4} \cdot \frac{1}{6} u^6 + C \\
 &= \frac{1}{24} u^6 + C \\
 &= \frac{1}{24} \cdot (2+x^4)^6 + C
 \end{aligned}$$

$$\begin{aligned}
 16). \int \cos^3 \theta \sin \theta \, d\theta & \quad \begin{array}{l} u = \cos \theta \\ du = -\sin \theta \, d\theta \end{array} \\
 &= -\int u^3 \, du \\
 &= -\frac{1}{4} u^4 + C \\
 &= -\frac{1}{4} (\cos^4 \theta) + C
 \end{aligned}$$

$$17). \int \frac{(\ln x)^2}{x} dx \quad \begin{array}{l} u = \ln x \\ du = \frac{1}{x} dx \end{array}$$

$$= \int u^2 du$$

$$= \frac{1}{3} u^3 + C$$

$$= \frac{1}{3} (\ln x)^3 + C$$

$$18). \int_0^{1/2} \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$$

$$= \int_0^{\pi/6} u du$$

$$= \frac{1}{2} u^2 \Big|_0^{\pi/6}$$

$$= \frac{1}{2} \left(\frac{\pi}{6} \right)^2 - \frac{1}{2} (0)^2$$

$$= \frac{1}{2} \cdot \frac{\pi^2}{36} = \frac{1}{72} \pi^2.$$

$$u = \sin^{-1} x$$

$$du = \frac{1}{\sqrt{1-x^2}} dx$$

$$\text{when } x=0, \quad u = \sin^{-1} 0 = 0$$

$$x=1/2, \quad u = \sin^{-1}(\frac{1}{2}) = \frac{\pi}{6}$$